

Innovation by Faculty

Dr. C. M. Ramakritinan

Dr. C.M. Ramakritinan has made significant contributions in the fields of Aquatic Ecotoxicology & Pollution, including ocean acidification, Marine Biotechnology, Marine Microbiology, Marine Natural Products, Coral Reef Ecology, Marine Molecular Taxonomy, and Socioeconomics over the past two and half decades. Recently, he secured a major research project from the Ministry of Earth Sciences (MoES) under Deep Ocean Mission to investigate the invasion through ship's ballast water and its Ecological Impacts," adopting an innovative approach to address critical threats in the maritime industry.

Key Innovations:

- Developed Seawater quality criteria for metals to the coral reef ecosystem of Gulf of Mannar. India (2003-2013).
- First time in India, a systematic study on Clade, sub-Clade and species level identification of coral endosymbionts of Palk Bay Reefs have been done and exclusively 139 sequences were published through NCBI using multilocus genotyping sequences. This study revealed thermotolerant endosymbionts in the Palk Bay corals (2014-2021).
- As first time, the corals of Palk Bay sequenced and obtained nine GenBank accession numbers using COI genes. This study opens a new path, for achieving monitoring and conservation of corals by using Potent Molecular Markers.
- Assessed Marine Micro and Macro Litters in the Coastal and Coral Reef Environment of Gulf of Mannar, a first systematic study in India (2011-2015).
- Reported 130 epibiotic species belonging to 52 families (fauna and marine flora including invasive bryozoan) along the whole Tamil Nadu Coasts (natural and artificial coastal defence structures).
- Studied the Impact of Ocean Acidification on Calcification of Marine Shellfishes of Gulf of Mannar using Ecotoxicological Testing (2015-2020).
- Assessed Coastal Pollution along the northern Gulf of Mannar using Microtox toxicity monitoring tool (2009-2012).
- Assessed the impacts of bleaching to the corals of Palk Bay during 1998, 2015 to 2020.
- Reported the impact of the tsunami of 26 Dec. 2004 on the 21 islands and corals of Gulf of Mannar and Palk Bay, India.
- Monitored the socioeconomic status of coral reef resource users of more than 20 fishing villages using Participatory Rural Appraisal (PRA) tools.

Dr. K. Altaff

Prof. K. Altaff, a distinguished researcher in live feed aquaculture, has an academic career spanning over four decades with pioneering contributions to copepod and cladoceran culture, invertebrate dynamics for environmental assessment, fish embryonic and larval development, and the use of copepods as live feed. His research has been supported by major grants from government agencies, including the University Grants Commission (UGC), Department of Biotechnology (DBT), Indian Council of Agricultural Research (ICAR), and the Ministry of Environment and Forests.

Key innovations:

- First global reports on several copepod species, including major groups from the Chennai coastal region.
- First Indian report on subitaneous and diapausing eggs in planktonic copepods.
- Development of copepod culture technology for large-scale aquaculture applications, enhancing the larval production of economically important finfish species.
- Comprehensive taxonomic redescrptions contributing to global copepod biodiversity knowledge.

Dr. M. Jayaprakashvel

Dr. M. Jayaprakashvel specializes in Marine Microbiology, Bioactive Secondary Metabolites, Plant Pathology, and Organic Agriculture. With 20 years of research experience, his doctoral work focused on the biological control of rice sheath blight through microbial consortia. His recent innovative research explores biofilm and biofouling studies in marine environments.

Key Innovations:

- Development of microbial consortia for biocontrol applications in agriculture.
- Discovery of bioactive compounds from endophytic fungi and their role in disease suppression.
- Major research project on Biofouling in Marine Environments, supported by MoES and the International Association of Maritime Universities (IAMU) in collaboration with Liverpool John Moores University, UK.
- Development of fish amino acids for sustainable agriculture.

Dr. R. Muthezhilan

Dr. R. Muthezhilan is a leading researcher in Marine Mycology and Bioprospecting of Marine Organisms, with research funded by IAMU, Liverpool John Moores University,

and TNSCST. His work focuses on antimicrobial peptides, biopolymers, and marine-derived food colorants.

Key Innovations:

- Advancements in marine-derived biopolymers for industrial applications.
- Development of antimicrobial peptides from marine fungi for biomedical use.
- Innovative approaches to marine mycology and bioprospecting with a focus on sustainable solutions.

Dr. C. Selvaraj

Dr. C. Selvaraj specializes in bioinformatics and drug design, focusing on the development of inhibitors to combat microbial infections and cancer. He has been honored with the prestigious NIH-sponsored ORISE Fellowship and the EU-funded Marie Skłodowska-Curie Action Award. His research is dedicated to exploring molecular-level interactions between pathogens and humans, leveraging chemical biology and drug discovery approaches. Dr. Selvaraj employs theoretical and biophysical methods, utilizing advanced bioinformatics techniques complemented by experimental validation to advance his groundbreaking work.

Key Innovations:

- Designed and implemented an open-source code and tool for a pIC50 calculator, enabling efficient drug potency analysis.
- Pioneered an Artificial Intelligence-driven method for similarity-based drug screening, which was patented under the Intellectual Property of India.
- Innovated a novel composition for an irradiated vaccine targeting *Aeromonas* species, securing a patent under the Intellectual Property of Germany.

Dr. R. Vijayaraj

Dr. R. Vijayaraj specializes in live feed technology, bioactive compounds, and their medicinal applications, particularly in Pharmacology and Toxicology. He received a major research project from the DST-Anusandhan National Research Foundation (ANRF), Govt. of India to develop mass production technology for marine copepods for enhancing marine finfish seed production.

Key Innovations:

- Development of high-density marine copepod culture technology for aquaculture hatcheries.

- Discovery of a novel antidiabetic compound from marine microalgae and its inhibitory effects on penicillin-binding proteins of *Staphylococcus aureus*.

Application of in silico modeling to study marine bioactive compounds against the novel coronavirus.